Huawei Access Node JAY-ACC-60 Optical Transport Failure Resolution SOP

Node: JAY-ACC-60 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_8846

Description

The failure of the optical transport domain on the Huawei Access Node JAY-ACC-60 at the Jayanagar Metro Site can be attributed to the degradation of the SFP/XFP transceivers or fiber cuts, resulting in a loss of receive and transmit power. This is a result of the inherent mechanics of optical transport, where the quality of the optical signal is susceptible to interference and attenuation over long distances. In the context of the Huawei Access Node, the SFP/XFP transceivers are responsible for converting the electrical signal to an optical signal and vice versa. Any degradation or failure of these transceivers can lead to a loss of signal quality, resulting in the ALARM_8846 alarm code. The role of the Access Node in this failure domain is critical, as it is responsible for aggregating and switching traffic between the metro and core networks. The failure of the optical transport domain can have a cascading effect on the network, resulting in packet loss and increased latency.

Pre-requisite Checks

- check: Verify that the redundant link is operational and can take over traffic in case of a failure of the primary link, status: Completed

- check: Ensure that the remote hands have been dispatched to the Jayanagar Metro Site to assist with the resolution, status: Completed

- check: Verify that the power supply to the Access Node is stable and within the normal operating range, status: Completed

Potential Root Causes

Cause	Probability
Degradation of SFP/XFP transceivers	High
Fiber cuts or damage	Medium
Rx/Tx power loss due to optical signal attenuation	Low

Diagnostic Commands & Expected Outputs

```
Command: display transceiver info, Expected Output: Transceiver information, including SFP/XFP type, optical power, and temperature, Description: This command is used to display the transceiver information, including the type of SFP/XFP, optical power, and temperature. This will help identify any degradation or failure of the transceivers.

Command: diagnostic fiber port, Expected Output: Fiber port information, including optical power, signal quality, and noise level,
```

```
Description: This command is used to diagnose the fiber port,
including the optical power, signal quality, and noise level. This
will help identify any issues with the fiber itself or the connection
to the transceiver.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP transceiver, Command: replace transceiver 1/1/1, Description: This command is used to replace the SFP/XFP transceiver on the 1/1/1 port. This will require physical access to the Access Node and the transceiver itself.

Step 2: Step: Clean and inspect the fiber optic cable, Command: diagnostic fiber clean 1/1/1, Description: This command is used to clean and inspect the fiber optic cable on the 1/1/1 port. This will require physical access to the fiber optic cable and the cable itself.

Post-Check Verification

```
Command: display transceiver info, Expected Output: Transceiver
information, including SFP/XFP type, optical power, and temperature,
Description: This command is used to verify that the SFP/XFP
transceiver has been replaced or cleaned and inspected successfully.

Command: diagnostic fiber port, Expected Output: Fiber port
information, including optical power, signal quality, and noise level,
Description: This command is used to verify that the fiber optic cable
has been cleaned and inspected successfully.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP transceiver, Command: replace transceiver 1/1/1, Description: This command is used to restore the original SFP/XFP transceiver on the 1/1/1 port. This will require physical access to the Access Node and the transceiver itself.

- Step: Re-clean and re-inspect the fiber optic cable, Command: diagnostic fiber clean 1/1/1, Description: This command is used to re-clean and re-inspect the fiber optic cable on the 1/1/1 port. This will require physical access to the fiber optic cable and the cable itself.